Roll No.: DL.AI.UM AID 24140

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – October 2024

First Semester

Artificial Intelligence

23EEE103 Introduction to Electrical Engineering

Duration: Two hours | Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Apply circuit laws and computational techniques to analyze and optimize electrical circuits.
CO02	Simulate and analyze the transient behavior of DC circuits using computational tools.
CO03	Simplify complex circuits and relevant theorems using Thevenin and Norton equivalents, incorporating AI where applicable.
CO04	Provides a comprehensive view of AC circuits, equipping students with the necessary skills to work with power systems and electronic devices.

Answer all questions

1. Find the power supplied or absorbed by each component in the given circuit of Fig.1. [5 marks] [CO01],[BTL5]

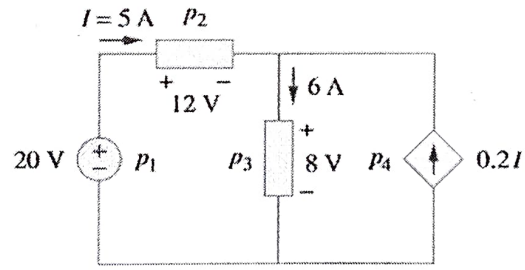


Fig. 1: Basic electrical circuit

2. A 1.5-kW toaster takes roughly 3.5 minutes to heat four slices of bread. Find the cost of operating the toaster once per day for 1 month (30 days). Assume energy costs 8.2 cents/kWh. [3 marks] [CO01], [BTL4]
3. Identify nodes, loops, and branches in the given Fig 2 circuit. [2 marks] [CO01], [BTL2]

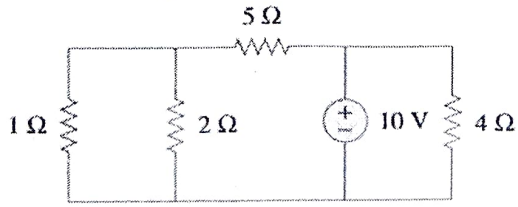


Fig. 2: Basic electrical circuit

R

4. Calculate the voltage v, the conductance G, and the power p for the Fig. 3. [3 marks] [CO02], [BTL3]

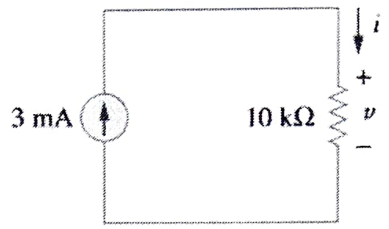


Fig. 3: Basic electrical circuit

5. Determine v_o and i in the given circuit of Fig. 4. [3 marks] [CO02], [BTL3]

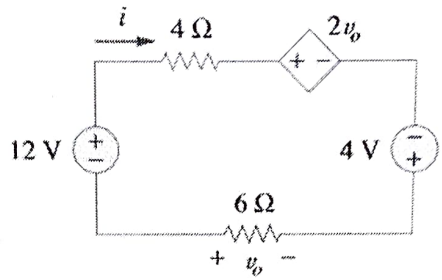


Fig. 4: Basic electrical circuit

6. Determine equivalent resistance for the given circuit of Fig. 5. [4 marks] [CO02], [BTL1]

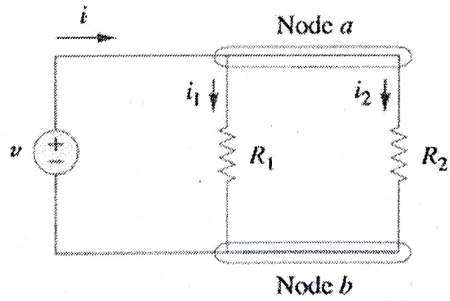


Fig. 5: Basic electrical circuit

7. Find equivalent resistance for the given circuit of Fig. 6. [5 marks] [CO02], [BTL6]

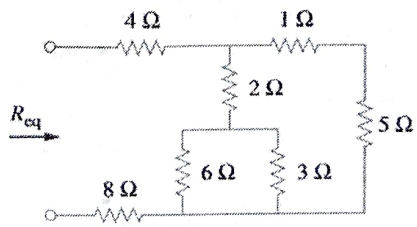


Fig. 6: Basic electrical circuit

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8. Calculate v_o, and i_o in the given circuit of Fig. 7. Also evaluate the power across 3 ohm resistor. [5 marks] [CO03], [BTL3]

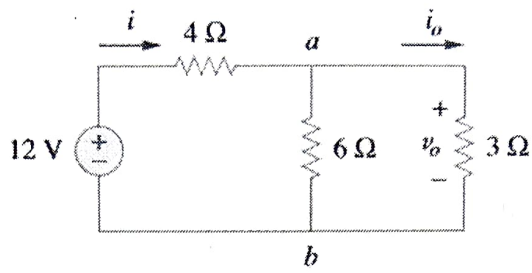


Fig. 7: Basic electrical circuit

9. Determine the node voltages across the given circuit of Fig. 8. [8 marks] [CO03], [BTL5]

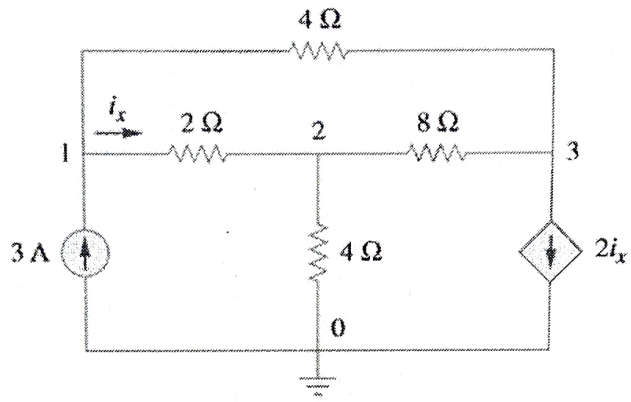


Fig. 8: Basic electrical circuit

10. Use mesh analysis to find the mesh currents in the given Fig. 9. [8 marks] [CO03], [BTL4]

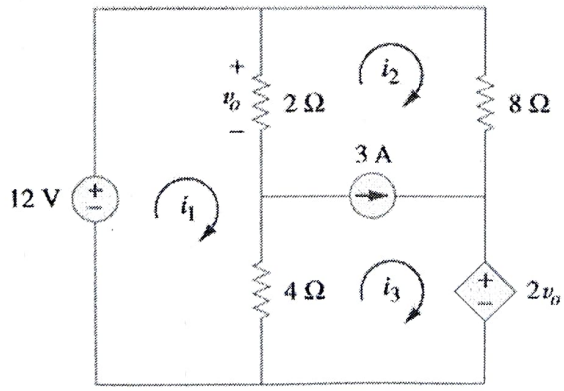


Fig. 9: Basic electrical circuit

11. Calculate node voltages in the circuit given in Fig. 10. [4 marks] [CO03], [BTL5]

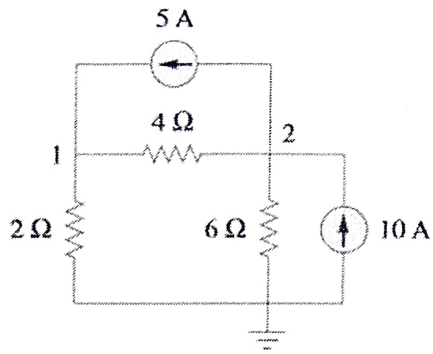


Fig. 10: Basic electrical circuit

Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**10**	BTL 1	**4**
CO02	**15**	BTL 2	**2**
CO03	**25**	BTL 3	**11**
CO04	**0**	BTL 4	**11**
CO05	**0**	BTL 5	**17**
CO06	**0**	BTL 6	**5**

Bloom's Taxonomy Levels (attached for reference)

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply

Level 4 - Analyze

Level 5 - Evaluate

Level 6- Create

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